FairPoC: Convergence-Stable Proof of Contribution for Auditable Federated Learning via ERC-20 Smart Contracts

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***Abstract*—Federated learning requires a mechanism to reward participants in proportion to their data contribution, but Shapley-based and leave-one-out metrics degrade once the global model converges, and existing systems distribute rewards through a central server with no public audit trail. We propose FairPoC, which scores contributions using normalised gradient norm, which is stable throughout training, and settles rewards through ERC-20 smart contracts with stake-and-slash enforcement: clients whose Proof-of-Contribution score falls below 30% of equal share are automatically slashed and blacklisted, with no trusted intermediary required for settlement. Across four non-IID regimes on CIFAR-10 and FashionMNIST, FairPoC achieves the highest or tied-highest accuracy in 7 of 8 configurations and the highest GT-Fairness in 4 of 8, with consistent gains over all baselines across the three moderate non-IID regimes on CIFAR-10. Under adversarial evaluation, free-rider attacks incur a net loss of $1.67\times$ the staked amount and gradient-inflation is capped below equal share by median-clip normalisation. Settlement costs $32.15 per round for 10 clients on Ethereum mainnet and about $0.32 on an L2 rollup, requiring no custom consensus and deploying on any EVM-compatible chain.**

***Index Terms*—federated learning, blockchain, proof of contribution, non-IID, ERC-20, smart contracts, stake-and-slash, adversarial robustness, fairness, gradient norm**

I. Introduction

Federated learning (FL) [1] allows multiple clients to train a shared model on their local data, with only model updates communicated to a central aggregator. This paradigm is attractive for privacy-sensitive domains such as healthcare, finance, and IoT [2]. However, FL introduces a fundamental economic problem: clients contribute heterogeneous amounts of value, yet standard aggregation schemes reward all participants uniformly. Recent surveys place this first among the unresolved problems in FL, describing incentive design as a foundational requirement for FL to operate at scale and one that must jointly address client heterogeneity, contribution quality, and fairness [3], [4]. Under non-IID data distributions (where clients hold skewed, non-representative class subsets), uniform reward policy over-rewards low-contribution clients and under-rewards those with rare, high-value data [5], [6].

The consequences are practical. A rational client that observes its scarce data is rewarded identically to a low-quality contributor will reduce participation or withdraw, degrading the federation [7]. Incentive mechanisms for FL have been studied [8], [9], but predominantly rely on a central server to distribute rewards with no externally verifiable record: participants must trust the server's claimed allocation, which is unauditable in cross-organisational deployments.

Blockchain technology offers *auditable* enforcement: once a reward distribution rule is encoded in a smart contract, execution is publicly verifiable and tamper-evident [10]. Behera et al. [11] show that Ethereum smart contracts can encode FL incentivization rules based on a scalar contribution derived from weight updates. BLADE-FL [12] integrates blockchain consensus into a decentralised FL aggregation framework. PoShapley-BCFL [13] encodes Shapley-based contribution into blockchain consensus. However, these works share a common limitation: Shapley values and leave-one-out (LOO) contributions collapse toward zero as the global model converges, making contribution measurement unreliable in late training stages [13]. Furthermore, existing scalar contribution metrics [11] are based solely on weight-update magnitude and do not account for distributional diversity, a key dimension of value under non-IID conditions.

Recent work proposes multidimensional contribution scores combining reputation, data size, gradient diversity (Jensen-Shannon divergence), and model performance, demonstrating improved fairness on heterogeneous FL tasks [8], [9]. However, such approaches do not integrate with blockchain for tamper-evident enforcement and do not address convergence-stability of diversity signals.

We address these gaps with **FairPoC**:

1) **A contribution signal that stays discriminative at convergence.** We score clients by normalised gradient norm, a heterogeneity-driven quantity that does not vanish once the global model saturates. Marginal-utility metrics (Shapley, leave-one-out) lose the ability to separate clients at precisely the point where reward allocation

must still be enforced.

2) **A non-circular protocol for evaluating contribution fairness.** GT-Fairness measures alignment between *ground-truth* data quality and reward, rather than between reward and the system's own score. It is therefore distinct from dispersion measures such as the Gini coefficient of payouts [14], which are minimised by paying every client equally regardless of contribution.
3) **Gradient-magnitude scoring is a manipulable primitive, and median-clip normalisation corrects it.** We characterise the self-amplifying failure mode of max-norm scoring under gradient inflation, demonstrate the same failure in a published critical-period detector [15] that gates bonus payment on gradient norms, and show that median-clip bounds both.
4) **Practical on-chain enforcement.** Stake-and-slash through standard ERC-20 interfaces, requiring no custom consensus, with measured gas costs. We claim deployment practicality here rather than novelty: on-chain reward enforcement is already established [16], [17].

II. Related Work

A. Contribution Measurement in FL

Shapley-value approximation [7], [8], [18] is the standard for FL contribution scoring but has two practical problems: $O(2^n)$ exact computational complexity and convergence instability (marginal contributions collapse as the model saturates). Xu et al. [9] propose gradient-driven reward allocation: each client is rewarded proportionally to the inner product of its gradient with the collaborative model's final gradient, guaranteeing equitable distribution by construction. Their approach targets centralised collaborative ML with no blockchain component and relies on gradient direction rather than scale. FairPoC uses normalised gradient *norm* (scale), a simpler closed-form signal that is independent of the consensus gradient direction and therefore stable throughout training, whereas inner-product-based methods converge toward the mean gradient and can lose client differentiation at convergence. Unlike these approaches, which assume a trusted central coordinator for distribution, FairPoC encodes rewards in an ERC-20 contract, enabling auditable, tamper-evident distribution with no custom consensus.

B. Blockchain-Assisted FL

Kim et al. [19] first proposed recording FL model updates on blockchain for auditability. BLADE-FL [12] replaces the central aggregator with a blockchain-based consensus mechanism where clients compete to generate blocks containing verified model updates. PoCL [20] proposes a multi-winner blockchain consensus algorithm that redistributes proof-of-work computation toward FL training. PoShapley-BCFL [13] encodes Shapley-based contribution scores into a custom blockchain consensus. Behera et al. [11] use Ethereum smart contracts with a scalar contribution metric derived from weight-update magnitude.

FLock [17], [21] proposes on-chain stake-and-slash for FL defence via smart contracts, where validators forfeit deposits for submitting poisoned updates. FairPoC shares the enforcement philosophy but differs in purpose: FLock targets Byzantine poisoning defence with peer-to-peer model evaluation; FairPoC targets *contribution-proportional reward fairness* under non-IID data, pairing slash with a non-IID-aware PoC score and a non-circular GT-Fairness evaluation protocol. Relative to this line of work, FairPoC replaces Shapley values and weight-update scalars with a normalised gradient norm that remains discriminative at convergence, and deploys through standard ERC-20 interfaces rather than a custom consensus rule. Earlier schemes that place contribution evaluation on chain do so chiefly to remove reliance on a semi-trusted scoring server [22], which is a different objective from the one pursued here.

C. Reputation-Based Trust and Incentives in Blockchain FL

A parallel line of work scores clients by *trustworthiness* rather than by contribution. TFFL [16] builds a hybrid subjective-logic reputation from time-decayed counts of positive and negative interactions, weights it by data quality (Earth Mover's Distance), performance and reliability, and distributes rewards through six Solidity contracts using cooperative game theory with Nash, Pareto and core guarantees. TAFL [23] extends the same reputation machinery into a detection, probationary-isolation and gated-reintegration lifecycle, committing reward tuples to an auditable ledger. Both are non-IID aware and both enforce rewards on chain. FairPoC therefore does not claim novelty from that combination.

The distinction lies in what the score measures. Reputation in TFFL accumulates from whether an update was *flagged* by an attack detector, which answers whether a client is trustworthy. Contribution-proportional reward requires a different quantity, namely how much each client added. TFFL allocates that share by Shapley value, and this is where the signal fails. Marginal-contribution scores are driven by the improvement a client makes to global accuracy, and that improvement vanishes for every client simultaneously as the model saturates, so the scores lose their ability to discriminate exactly when the model is most valuable. The degeneracy is already acknowledged in the Shapley literature, though not named as such: GTG-Shapley [24] discards any round whose total marginal gain falls below a threshold and returns *zero* for every participant in it, and Geimer et al. [25] report that Shapley-based allocations are unstable across aggregation rules. Figure 2 quantifies the same effect over training time: by round 30 of 50 the leave-one-out signal has collapsed to near zero for every client, while the normalised gradient norm remains discriminative throughout.

This matters beyond reward accounting. Establishing theoretical guarantees for adaptive, context-aware client selection is named as an open direction in the FL literature [3], and any such guarantee must rest on a signal that still separates clients at the round in which the selection or payment is made.

Algorithm 1: FairPoC Training Protocol

Input: Global model $\theta^{(0)}$, rounds T, clients $\{1 \dots n\}$
for $t = 1$ **to** T **do**
  Aggregator broadcasts $\theta^{(t)}$ to all clients;
  Each client i trains locally, produces Δw_i;
  Clients send Δw_i to aggregator;
  Aggregator computes PoC scores s_i (Eq. 1);
  Aggregator calls `submitAndSettle(`s_i`)` on-chain;
  Contract mints and transfers tokens $\propto s_i$;
  $\theta^{(t+1)} \leftarrow \sum_i \frac{s_i}{\sum_j s_j} \theta_i^{(t)}$;

FairPoC's contribution is thus not the pairing of non-IID-aware scoring with on-chain enforcement, which TFFL already provides, but a contribution signal that remains non-degenerate at the point where reward allocation and slashing must still be enforceable.

III. System Design

A. Architecture Overview

FairPoC has three roles: *clients*, an *aggregator*, and the *FairPoC smart contract*. Algorithm 1 summarises the per-round protocol.

The aggregator computes contribution scores and writes them to the contract via `submitAndSettle()`. The smart contract then executes reward distribution deterministically and immutably: once scores are posted, no party, including the aggregator, can alter the resulting token transfer. This makes the distribution step *auditable and tamper-evident*: any participant can verify every reward on-chain. Note that score computation itself remains at the aggregator; full trustlessness of scoring would require on-chain gradient verification, which we identify as future work.

B. Smart Contract Design

`FairPoC.sol` extends ERC-20 [26] with three on-chain functions encoding both reward distribution and adversarial defence. `stake()` locks a per-round deposit before training begins. `submitAndSettle()` is called once per round by the aggregator: for each client it atomically checks whether the PoC score exceeds 30% of equal share (a threshold that tolerates mild underperformance while decisively penalising near-zero free-rider scores; sensitivity sweep is future work). Clients above receive proportional reward plus any slashed tokens; clients below are blacklisted and their stake is forfeited to the slash pool. Because settlement executes atomically on-chain, no trusted party can override the slashing decision. `isBlacklisted()` provides a public audit hook. Using ERC-20 as the base makes tokens compatible with any EVM wallet.

IV. Non-IID-Aware Proof of Contribution

A. PoC Score

Under non-IID data, a client's value has two components: how accurate its local model is, and how much distributional diversity it contributes. We define:

$$s_i = \alpha_{\mathrm{p}} \cdot \mathrm{ValAcc}_i + \beta_{\mathrm{p}} \cdot \mathrm{GradNorm}_i \tag{1}$$

with $\alpha_{\mathrm{p}} = 0.7$ and $\beta_{\mathrm{p}} = 0.3$. ValAcc_i is the top-1 accuracy of client i's local model on the shared validation set; $\mathrm{GradNorm}_i \in [0, 1]$ is defined in Section IV-B.

B. Gradient Norm Diversity Signal

$\mathrm{GradNorm}_i$ measures the magnitude of client i's parameter update relative to the other clients in the same round:

$$\mathrm{GradNorm}_i = \frac{\|\Delta w_i\|_2}{\max_j \|\Delta w_j\|_2} \tag{2}$$

mapping values to $[0, 1]$ within each round. Clients holding rare or diverse class subsets produce larger parameter updates because their local loss surface diverges more from the current global model direction. Under moderate non-IID partitioning, gradient magnitude is positively correlated with data volume and distributional diversity. This holds while clients retain several classes; Proposition 1 shows that the correlation with volume breaks down as class concentration $\rho_i \to 1$, and Corollary 1.1 traces the resulting fairness loss. The regime boundary is visible in our results: the correlation supports FairPoC at $\alpha_D \geq 0.5$, and fails at $\alpha_D = 0.1$ on FashionMNIST.

Unlike Shapley/LOO contributions, which measure marginal accuracy improvements and collapse to zero once the model saturates, $\mathrm{GradNorm}$ reflects client i's local data distribution and remains non-degenerate throughout training, directly addressing the limitation acknowledged by PoShapley-BCFL [13].

C. Byzantine Filtering

FairPoC applies a soft statistical filter before aggregation: clients with $s_i < \mu_s - 2\sigma_s$ are excluded, where μ_s and σ_s are the round-level mean and standard deviation of scores.

V. Theoretical Analysis

A. Smart Contract Distribution Integrity

Let $s_i > 0$ be client i's PoC score, $S = \sum_i s_i$, and R the per-round token pool.

Remark 1 (Contract Integrity Property). The immutable `FairPoC` contract executes $r_i = (s_i/S) \cdot R$ on-chain, guaranteeing exact proportionality to submitted scores: $r_i/s_i = R/S$ is identical for all i. This is a *distribution faithfulness* guarantee: the on-chain function cannot be silently modified after deployment, making every reward distribution publicly auditable and tamper-evident. Note that score computation itself remains off-chain at the aggregator; the contract enforces faithfulness to *whatever scores are submitted*, enabling retrospective dispute resolution via on-chain logs.

B. Gradient Norm Stability

Remark 2 (Gradient Norm Stability (empirical)). Per-client gradient norms remain client-ordered and bounded away from zero throughout all 50 rounds at $\alpha_D = 0.1$ (Fig. 2), while LOO margins collapse after round 30 as global accuracy saturates. Since $\|\Delta w_i\|$ depends on client i's local loss landscape and not on other clients' gradients, it remains non-degenerate as long as any local distribution differs from the global model.

C. Gradient Norm Decoupling at Extreme Non-IID

Proposition 1 (Gradient Norm Decouples from Data Volume)**.** *Let $\rho_i = \max_k p_{i,k}$ be client i's class concentration. Assume cross-entropy loss, $E \geq 1$ local SGD epochs, and $n_i \geq B$. Then $\|\Delta w_i\| \approx \rho_i \cdot \|g_{k_i^*}(w)\|$, where $k_i^* = \arg\max_k p_{i,k}$. In particular, $\|\Delta w_i\|$ is independent of n_i up to the factor ρ_i.*

Proof sketch. Under extreme non-IID ($\rho_i \to 1$), the local loss is dominated by class k_i^*. After E local SGD epochs the update converges to $\Delta w_i \approx \eta_{\text{eff}} \rho_i g_{k_i^*}(w)$, where $g_{k_i^*}$ is the population gradient for that class, independent of n_i provided $n_i \geq B$ (one mini-batch). Hence $\|\Delta w_i\| \approx \eta_{\text{eff}} \rho_i \|g_{k_i^*}(w)\|$ is independent of n_i up to ρ_i. □

Corollary 1.1 (GT-Fairness Degrades at $\alpha_D \ll 1$)**.** *The GT ground truth satisfies $\text{gt}_i \approx \gamma(n_i/N)$ when $\rho_i \to 1$ (entropy $H_i \to 0$). Hence $r_i/\text{gt}_i \propto 1/n_i$, which varies across clients, causing $\mathcal{J}(r_i/\text{gt}_i) < 1$.*

Corollary 1.1 predicts exactly the FashionMNIST α_D=0.1 result in Table II: all clients are near-single-class, gradient norms reflect the dominant-class gradient rather than data volume, and smaller clients are over-rewarded relative to their GT weight. At $\alpha_D \geq 0.5$, clients hold multiple classes ($\rho_i < 1$, $H_i > 0$), and gradient norm correlates with data volume through the multi-class sum; FairPoC leads all baselines on both datasets.

VI. Experiments

A. Setup

We evaluate on CIFAR-10 [27] (ResNet-8) and Fashion-MNIST [28] (3-layer CNN), partitioned across 10 clients via Dirichlet(α_D) with $\alpha_D \in \{0.1, 0.5, 1.0, 10.0\}$; lower values indicate more extreme heterogeneity (Fig. 1, α_D=0.1). Training runs for 50 rounds, 3 local epochs, batch size 32, learning rate 10^{-3}; results are means over seeds {42, 123, 456}. Baselines are:

- **FedAVG** [1]: uniform aggregation.
- **PoCL** [20]: accuracy-proportional scoring, low-accuracy clients filtered at $\mu - 2\sigma$, no blockchain.
- **BLADE-FL** [12]: cosine-similarity Byzantine filter at $\mu - 2\sigma$; retained clients get uniform weights.
- **TMC-Shapley** [29]: truncated MC Shapley, T=20 permutations, negatives clipped to zero.

We report two metrics. Accuracy is the final-round top-1 on the full test set. GT-Fairness is Jain's index [30] of reward-to-ground-truth ratios averaged over all rounds, where the ground truth for client i is

$$\text{gt}_i = 0.5\,\tfrac{n_i}{N} \;+\; 0.5\,H_i, \tag{3}$$

with H_i the normalised Shannon entropy of client i's class proportions and r_i its share of the round reward pool: GT-Fairness $= \mathcal{J}(r_i/\text{gt}_i)$. Because gt_i is derived solely from the data partition, the metric is non-circular and independent of any scoring method.

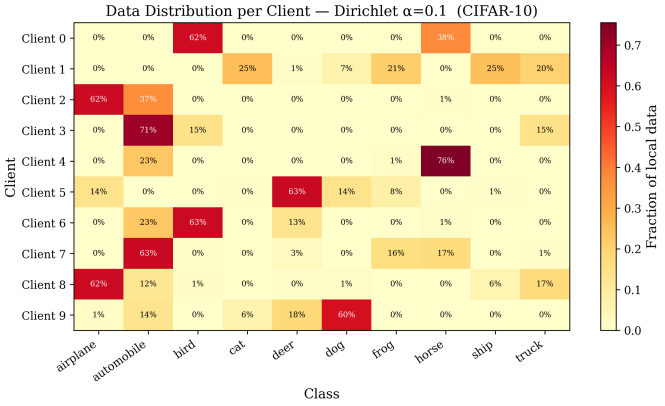


Fig. 1. Class distribution across 10 clients at $\alpha_D = 0.1$. Rows are clients; columns are classes.

TABLE I
CIFAR-10: Final Accuracy (%) / GT-Fairness (3-seed mean). Bold = best per column.

Method	α_D=0.1	α_D=0.5	α_D=1.0	α_D=10.0
FedAVG	73.0 / 0.904	82.2 / 0.981	84.2 / 0.993	**85.7** / **1.000**
PoCL	73.1 / 0.879	82.5 / 0.989	84.2 / 0.929	85.5 / 0.933
FLTrust	59.9 / 0.500	76.3 / 0.530	80.9 / 0.500	85.5 / 0.730
FairPoC	**74.9** / **0.969**	**82.9** / **0.994**	**84.3** / **0.997**	**85.7** / 0.967

B. Main Results

Tables I and II report final accuracy and average Jain's fairness.

On CIFAR-10, FairPoC achieves the highest GT-Fairness at $\alpha_D \in \{0.1, 0.5, 1.0\}$, with the largest gap at $\alpha_D = 0.1$ (0.969 vs. FedAVG 0.904); PoCL falls further to 0.879 because accuracy-only weighting ignores contribution diversity. At $\alpha_D = 10.0$ (near-IID), FedAVG achieves the maximum of 1.000 because uniform rewards are trivially optimal when all clients hold balanced data; FairPoC reaches 0.967 in this regime. FLTrust sacrifices fairness for Byzantine robustness, falling to 0.500 at $\alpha_D \in \{0.1, 1.0\}$. On FashionMNIST, FairPoC leads at $\alpha_D = 0.5$ (0.991 vs. FedAVG 0.976) and ties at $\alpha_D = 10.0$, but falls below FedAVG at $\alpha_D = 0.1$ (0.595 vs. 0.649) and below PoCL at $\alpha_D = 1.0$ (0.962 vs. 0.995). The $\alpha_D = 0.1$ shortfall arises because near-single-class clients produce similarly scaled gradient norms regardless of data volume, causing the PoC score to misalign with the volume-weighted GT (see §VII). Accuracy is consistently

TABLE II
FASHIONMNIST: FINAL ACCURACY (%) / GT-FAIRNESS (3-SEED MEAN). BOLD = BEST PER COLUMN. †: FAIRPOC FALLS BELOW FEDAVG AT EXTREME NON-IID (SEE TEXT).

Method	α_D=0.1	α_D=0.5	α_D=1.0	α_D=10.0
FedAVG	**89.4** / **0.649**	92.5 / 0.976	**92.8** / 0.994	93.2 / **1.000**
PoCL	88.9 / 0.515	92.4 / 0.949	92.7 / **0.995**	**93.3** / **1.000**
FLTrust	78.7 / 0.337	90.8 / 0.487	92.0 / 0.549	93.0 / 0.504
FairPoC	89.3 / 0.595†	**92.6** / **0.991**	**92.8** / 0.962	**93.4** / **1.000**

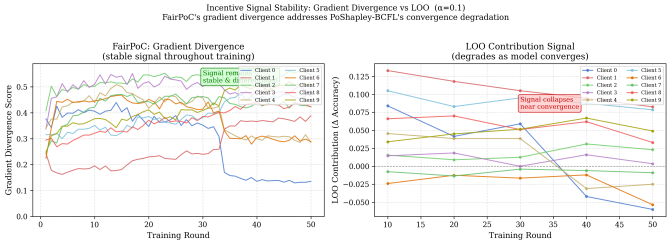


Fig. 2. Gradient norm diversity (left) vs. LOO (right) over 50 rounds at $\alpha_D = 0.1$. Gradient norm remains stable and client-differentiated throughout; LOO collapses toward zero after round 30, consistent with the limitation acknowledged by PoShapley-BCFL [13].

competitive: FairPoC leads or ties on CIFAR-10 in all settings, and differences are within 0.2% on FashionMNIST.

C. Convergence Stability of Gradient Divergence

Fig. 2 compares gradient norm and LOO over 50 rounds at $\alpha_D = 0.1$. LOO collapses after round 30 as accuracy saturates, confirming the instability acknowledged by PoShapley-BCFL [13]. Gradient norm remains stable and client-ordered throughout, which is a necessary condition for reliable long-run contribution scoring.

D. Adversarial Robustness

A key limitation of purely off-chain FL defences is that they rely on a trusted aggregator: a colluding aggregator can override filtering and over-reward an attacker. FairPoC addresses this at the protocol level via a *stake-and-slash* mechanism encoded in the smart contract (`FairPoC.sol`): each client locks a fixed stake before each round, and the contract autonomously slashes and blacklists any client whose on-chain PoC score falls below 30% of the equal-share baseline, redistributing the forfeited stake to honest clients in the same round. No trusted third party *executes the slashing decision*: the contract code enforces the rule immutably and transparently.

We evaluate three representative attacks at $\alpha_D = 0.1$ (CIFAR-10, one attacker out of 10 clients):

- **GradScale**: update scaled by $10\times$ to inflate GradNorm.
- **FreeRider**: submits global model unchanged (zero update).
- **LabelFlip**: trains on labels flipped to $(C-1-y)$.

Table III reports accuracy, GT-Fairness, and net-gain (reward earned minus stake lost relative to fair share; values below $1.0\times$ make attacking economically irrational).

TABLE III
ADVERSARIAL ROBUSTNESS (α_D=0.1, CIFAR-10, 1 ATTACKER / 10 CLIENTS). OVER-REWARD = ATTACKER REWARD SHARE $\div\, 1/n$. NET-GAIN <1.0×: ECONOMICALLY IRRATIONAL TO ATTACK. BOLD = BEST PER ATTACK. FAIRPOC-MC USES MEDIAN-CLIP.

Attack	**Method**	**Acc (%)**	**GT-Fair**	**Over-rwd**	**Net-gain**
GradScale	FedAVG	55.54	0.904	1.00×	–
	BLADE-FL	59.69	0.901	1.01×	–
	FairPoC	17.83	0.932	1.42×	1.42×
	FairPoC-MC	**68.09**	**0.965**	**0.78**×	**0.78**×
FreeRider	FedAVG	70.11	0.904	1.00×	–
	BLADE-FL	69.80	0.904	0.98×	–
	FairPoC	71.71	**0.968**	1.01×	1.01×
	FairPoC-MC	**72.02**	0.856	**0.00**×	**−1.67**×
LabelFlip	FedAVG	73.25	0.904	1.00×	–
	BLADE-FL	72.26	0.904	1.00×	–
	FairPoC	**74.64**	**0.974**	**0.88**×	**0.88**×
	FairPoC-MC	74.33	0.960	0.88×	0.88×

Manipulating a contribution metric in order to inflate one's own share of the reward pool is an established attack class [31], [32]. Our concern is narrower: the specific form this takes when the metric is a gradient magnitude, and what removes it.

Under GradScale, max-norm scoring is self-amplifying: the attacker becomes the normalisation maximum and suppresses honest clients to GradNorm≈ 0.1, collapsing accuracy to 17.83%, worse than FedAVG. FairPoC-MC breaks this by clipping the attacker to the median, recovering 68.09% accuracy at over-reward $0.78\times$. For free-riders, zero gradient norm yields a score of $\alpha_p \cdot \mathrm{ValAcc}_{\mathrm{global}}$ which may exceed honest clients in late rounds; FairPoC-MC slashes the attacker every round (net-gain $-1.67\times$). The resulting GT-Fairness of 0.856 is a metric artefact (the slashed client holds r_i=0 while counting in Jain's denominator) and does not reflect honest-client fairness. Under LabelFlip, near-chance ValAcc triggers blacklisting; both FairPoC variants under-reward the attacker ($0.88\times$) while achieving the highest accuracy (74.6%). FashionMNIST results are consistent: free-riders are slashed on both variants (net-gain $-1.67\times$) and FairPoC-MC leads on GradScale accuracy (89.1%). LabelFlip shows mild over-reward ($1.12\times$) because at $\alpha_D = 0.1$ honest clients also score low on the shared validation set, a known limitation of accuracy-based scoring under severe data skew.

E. Beyond FairPoC: Manipulating a Published CLP Detector

The GradScale failure mode is not specific to FairPoC. Any mechanism that aggregates gradient magnitudes inherits it, including schedules that gate payment on a detector built from those magnitudes. We demonstrate this on R3T [15], a contract-theoretic incentive framework that pays a bonus multiplier $h(t) = 1+\vartheta/\ln(2t)$ during *critical learning periods* and $h(t) = 1$ otherwise. R3T identifies those periods from a

TABLE IV
MANIPULATING THE R3T CRITICAL-PERIOD DETECTOR (CIFAR-10, α_D=0.1, 40 ROUNDS, 1 ATTACKER / 10 CLIENTS, τ=0.01). DRAIN IS BONUS BUDGET RELATIVE TO THE ATTACK-FREE ARM.

Arm	CLP rnds	Drain	Peak scale	Acc (%)
No attacker	2	1.000×	1.0	**69.30**
Fixed scale	1	0.995×	2.0	73.70
Ramped scale	32	1.208×	115	9.65
Adaptive	**26**	**1.182**×	**9.3**	54.25
Adaptive + median-clip	9	1.042×	1.0	70.15

federated gradient norm,

$$\mathrm{FGN}(t) = -\eta \sum_{n \in S(t)} \omega_n \, \|g_{B,n}\|_2^2, \qquad \omega_n = e_n / \textstyle\sum_m e_m, \tag{4}$$

classifying round t as critical when the *relative* change $(\mathrm{FGN}(t) - \mathrm{FGN}(t{-}1))/\mathrm{FGN}(t{-}1) \geq \tau$. Because the test is on the relative change, a client that raises its own term each round can hold the window open past the true critical phase. R3T assumes clients report truthfully and lists adversarial behaviour in critical periods as future work.

We implement Eq. 4 with $\tau = 0.01$ and run five arms on CIFAR-10 at $\alpha_D = 0.1$ over 40 rounds, one attacker in ten clients. *Fixed* scales by a constant, *ramp* by a geometrically growing factor, and *adaptive* solves each round for the *minimum* scale that keeps the round classified critical, so the reported inflation is the attack's true cost rather than a tuned value. Results are in Table IV.

Three observations follow. First, the *fixed* arm is a control and behaves as predicted: a constant inflation shifts FGN once and produces *fewer* critical rounds than the attack-free arm, confirming that the effect comes from ramping rather than from inflation as such. Second, the adaptive attacker extends the window from 2 rounds to 26 and inflates the bonus budget by 18.2% at a peak inflation of only 9.3×. Third, median-clip normalisation, the same correction that bounds GradScale in Section VI-D, cuts the window to 9 rounds and the drain to 4.2%, and restores accuracy.

Separately, the detector is marginal even with no adversary present. In the attack-free arm the round-to-round relative change of FGN ranges from -0.0175 to $+0.0117$ after round 10, so it straddles $\tau = 0.01$ in both directions and triggers a false critical round at round 40. The threshold sits inside the signal's own noise band.

Four limitations bound these results. The attack is not stealthy: it costs 15.0 points of final accuracy, which an operator would notice. The attacker gains no *share* advantage, since its bonus weight is fixed by data volume; it inflates the pool for everyone, so this is budget drain against the operator rather than theft from peers. We do not model R3T's budget feasibility constraint, under which the real effect would be early budget exhaustion or diluted per-client rewards rather than overspend. Finally, these are single-seed results at one heterogeneity level, and the noise-floor observation in particular warrants replication.

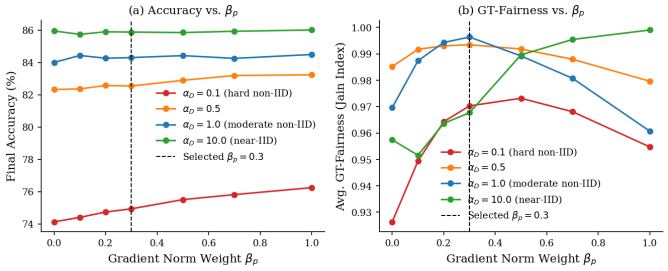


Fig. 3. Accuracy and GT-Fairness as β_p (gradient norm weight) varies from 0 (accuracy-only) to 1 (gradient-norm-only) on CIFAR-10. Dashed line: paper default $\beta_p = 0.3$.

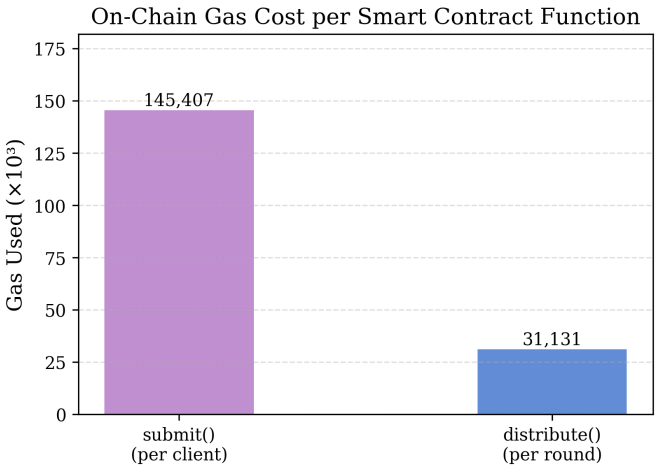


Fig. 4. Gas cost per smart contract call (Ganache testnet, single-client batch). `submitAndSettle([1 client])`: 145,407 gas (base scoring + reward); stake-and-slash adds ≈8,000 gas per client for the threshold check and slash accounting.

F. Sensitivity to β

Fig. 3 sweeps $\beta_p \in \{0.0, 0.1, \ldots, 1.0\}$ at $\alpha_D \in \{0.1, 0.5, 1.0, 10.0\}$ on CIFAR-10. At $\alpha_D \geq 0.5$, fairness peaks at $\beta_p \approx 0.3$ while accuracy is flat. At hard non-IID ($\alpha_D = 0.1$), accuracy rises monotonically but $\beta_p = 0.3$ captures most of the fairness gain before the decline ($< 1\%$ accuracy gap vs. $\beta_p = 1$). Higher β_p also weakens label-flip protection since divergent updates gain larger norm reward. $\beta_p = 0.3$ therefore balances fairness, accuracy, and robustness.

G. Blockchain Overhead

We deploy `FairPoC.sol` to a Ganache testnet and measure gas directly. The batched `submitAndSettle()` accepts all n scores in one transaction. A least-squares fit over $n \in \{10, 25, 50, 100, 200\}$ gives:

$$G(n) = 77{,}344 + 24{,}609\, n, \tag{5}$$

with $R^2 = 0.9998$, where 77,344 is the fixed per-round overhead (EVM base cost, threshold computation and reward-pool accounting) and 24,609 is the per-client marginal cost of classification, slashing and reward transfer. Deployment costs 1,403,789 gas once, and `stake()` costs 52,656 gas per client

TABLE V
MEASURED ON-CHAIN GAS COST OF FAIRPOC.SOL SUBMITANDSETTLE() ON A GANACHE TESTNET VS. NUMBER OF FL CLIENTS. *PROJECTED FROM EQ. 5. STAKE() ADDS 52,656 GAS PER CLIENT PER ROUND.

Clients	Total gas	ETH (30 Gwei)	USD ($3k/ETH)
10	357,249	0.0107	$32.15
50	1,274,483	0.0382	$114.70
100	2,522,270	0.0757	$227.00
500*	12,381,844	0.3715	$1,114.37
L2 (Optimism): $\approx 100\times$ cheaper than above			

per round. Table V reports costs; at 30 Gwei, ETH = $3,000, a 10-client round costs $32.15 on Ethereum mainnet, or about $0.32 on an L2 rollup.

VII. LIMITATIONS

All experiments run on a local Ganache testnet; gas price variability and public-chain latency are not captured. Score computation stays at the aggregator; on-chain gradient verification for fully trustless scoring is left to future work. The $n{=}10$ client setting is representative of private enterprise FL; Table V projects $O(n)$ cost to 500 clients, but accuracy at larger n is not studied.

At $\alpha_D = 0.1$ on FashionMNIST, FairPoC's GT-Fairness (0.595) falls below FedAVG (0.649). Under near-single-class concentration, gradient norms reflect the dominant class's magnitude rather than data volume (Proposition 1), so scores misalign with the volume-weighted GT. This effect is confined to degenerate concentration; at all $\alpha_D \geq 0.5$ FairPoC leads or matches every baseline on both datasets.

Stake-and-slash deters rational attackers but does not cryptographically exclude them: a patient adversary can build honest history until staking costs are recovered, then attack. An attacker who knows the median-clip rule cannot reliably exploit it (norms are posted before the round's median is published, making precise targeting a fixed-point problem with no advance solution), and any GradScale factor $k{>}1$ degrades validation accuracy, further lowering the attacker's PoC score. Zero-knowledge gradient proofs would close this gap and are future work.

VIII. CONCLUSION

We presented FairPoC, a blockchain-assisted FL framework that enforces non-IID-aware contribution rewards via ERC-20 smart contracts. Normalised gradient norm replaces Shapley/LOO signals that collapse at convergence; stake-and-slash provides empirically demonstrated economic deterrence for rational attackers, extending the enforcement principle of FLock to the contribution-fairness setting. Evaluation on CIFAR-10 and FashionMNIST confirms FairPoC achieves the highest or tied-highest accuracy in 7 of 8 configurations and the highest GT-Fairness in 4 of 8, with consistent fairness gains across moderate non-IID regimes on CIFAR-10, deployable on any EVM-compatible chain without custom consensus.

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